

Statistical Shot Quality Weighting

An Investigation of NHL Shot Quality and Offensive Performance

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Introduction

The following is an attempt to understand the significance of shot quality and shot execution in relation to the offensive performance of NHL teams and individual players by examining their shooting distribution. This was accomplished with analysis of the play-by-play statistics of 2 NHL seasons, where the specific goal scoring probabilities for a number of different game situations were calculated. Further to the research and data presented, several opposing interpretations of the results are discussed with the interest of encouraging further research and debate.

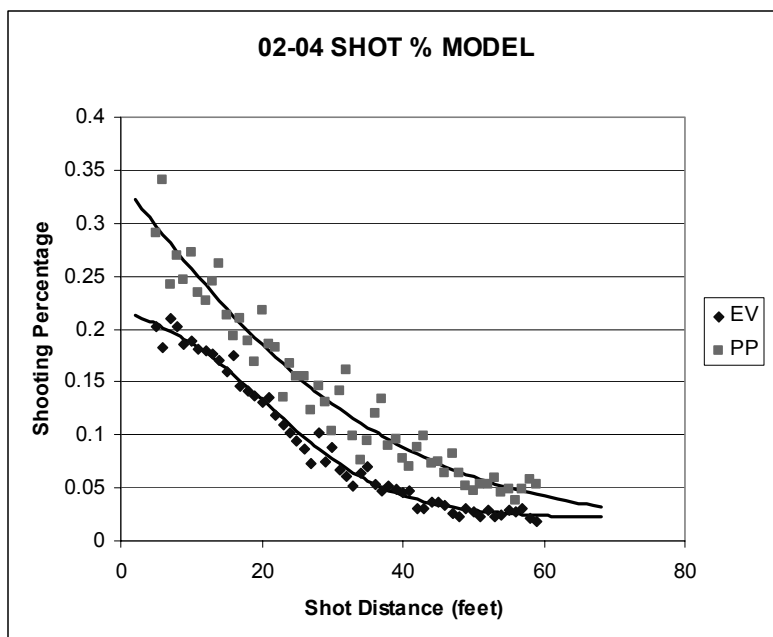
Background

Oddly, this study began as an inquiry into the nature of goaltending and defence. After I had worked on the topic for a while, I found out that Alan Ryder had completed this task already, using a method very similar to the one outlined below. Using the same play-by-play statistics, he defines a “shot quality neutral save percentage” to better evaluate the performance of goaltenders independent of the defence they have to play behind. I will focus mainly on offensive analysis, but refer to Alan’s goaltending and defence studies for excellent analysis using a similar method.

What I have done is separate every shot and goal by team and shooter. This gives me a season long record of every shot and goal, for and against, subdivided by shot distance in feet (0 feet to 59 feet, then every shot greater than 60 feet is grouped into a single “behind the blue-line” group).

First I attempted to understand the relationship between shot distance and scoring probability. I would definitely expect the probability of scoring to increase the closer you get to the goal, but is the relationship linear?

Exponential? To find out, I plotted the league-wide shooting percentage from all distances in my groups.



As the graph above demonstrates, logistic functions form the best fit with the data. The shots were subdivided by situation, namely power play or even strength, since their respective distributions are significantly different. I also compensated for empty net goals. Alan Ryder’s method is a bit more sophisticated in this regard. He has accounted for rebounds and shot type (backhand, slap, snap wrist, etc) in formulating his scoring probability model. I assumed the difference made by neglecting these extra factors is negligible for the purposes of this study. I found the data became too noisy when dealing with such small subdivisions for them to be of use. I welcome anyone to try a different method, and I would be very interested to see how well the results would agree with those that I have discovered.

Weighted Shots

If we accept this model as the league average, we can figure out a theoretical value for a team's and individual player's regular season goals for and against totals based on these formulae as they relate to their regular season shot distribution. By getting the computer to calculate the appropriate solution to the shot distribution logistic functions for every event, and then summing the event probabilities for every player and team, we can achieve total "theoretical goals for", or "theoretical goals against". All of this is solely from a shot distance and situation distribution.

I'll call theoretical goal totals "Weighted Shots For" and "Weighted Shots Against". The nice thing about these numbers is that they should approach the actual goal totals, allowing quick and intuitive comparison.

Example: 2002-2003 (truncated data) Mark Recchi
 Shots – 152
 Weighted Shots For (WSF) – 15.91
 Actual Goals Scored - 15

This is where the differences between the offensive and defensive analysis become complicated. In theory, every team should score, or be scored against, at a rate similar to the one predicted by the statistical model. Fortunately most teams do just that, however it turns out some teams deviate substantially from what this model would predict.

Brief Summary of Defensive and Goaltending Analysis

From the defensive side of the analysis, statistical anomalies do not present too much of a problem, because as Alan Ryder showed, you can assume that the discrepancy is a direct result of a wealth or detriment of goaltending ability. For example, here are some of my preliminary WSA defensive calculations for the 2003-2004 season:

TEAM	THEORETICAL GOALS AGAINST	ACTUAL GOALS AGAINST
Minnesota Wild	208.8	183
New York Rangers	241.9	250
Florida Panthers	235.4	221
New Jersey Devils	164.8	164

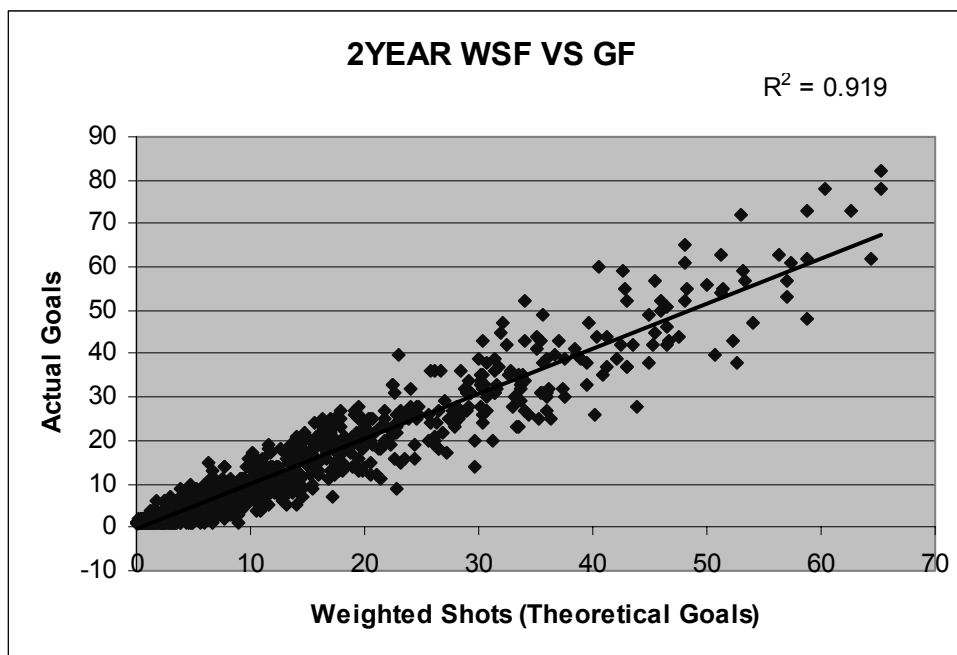
Three of the four teams deviate substantially from the theoretical value. It seems logical to conclude that Minnesota (Roloson and Fernandez) and Florida (Luongo) were the beneficiaries of elite goaltending. New York on the other hand, while appearing to be a dismal team defensively based solely on shot distribution, were also hindered by the collective inability of their goalies to stop shots at a league average rate.

One of the most interesting results was New Jersey and Martin Brodeur, who has a reputation as one of the best, if not *the* best NHL goaltender. Weighting for shot quality seems to expose his performance as

merely average, with his conventional statistics (and vezina trophy) artificially buoyed by his team's awesome defence. If SQN save percentage is a repeatable trait of an individual goaltender, the stat represents an extremely powerful forecasting and performance evaluation tool. Imagine the media reaction to trading Brodeur or Belfour for Dwayne Roloson!

Offensive Analysis

As exciting a prospect as defensive and goaltending evaluation is, offensive evaluation has proven to be much more frustrating. With no goaltender to give credit (or blame) for discrepancies, unexpected results have become the norm, with convincing explanations and conclusions much more rare. To illustrate the analysis, consider the results of the application of shot quality weighted scoring probabilities. On a player by player basis, the results are acceptable. Applying the shot probability model to individual players represents a much more accurate image of a player's performance than simply considering his shots on goal, especially considering a players performance over the full two year time span:



INDIVIDUAL PLAYER STUDY

LINEAR CORRELATION R^2

02-03 SHOTS VS GOALS	0.71
02-03 WEIGHTED SHOTS VS GOALS	0.85
02-04 WEIGHTED SHOTS VS GOALS	0.92

Fantastic. As expected, accounting for shot quality increases our understanding of a player substantially. For a trivial example, it shows convincingly why Vincent Lecavalier was a more potent goal scorer than Al MacInnis in 02-03, despite MacInnis getting more shots on goal (shots/goals scored/weighted shots):

Al MacInnis: 271 / 14 / 12.02
 Vincent Lecavalier: 263 / 30 / 31.62

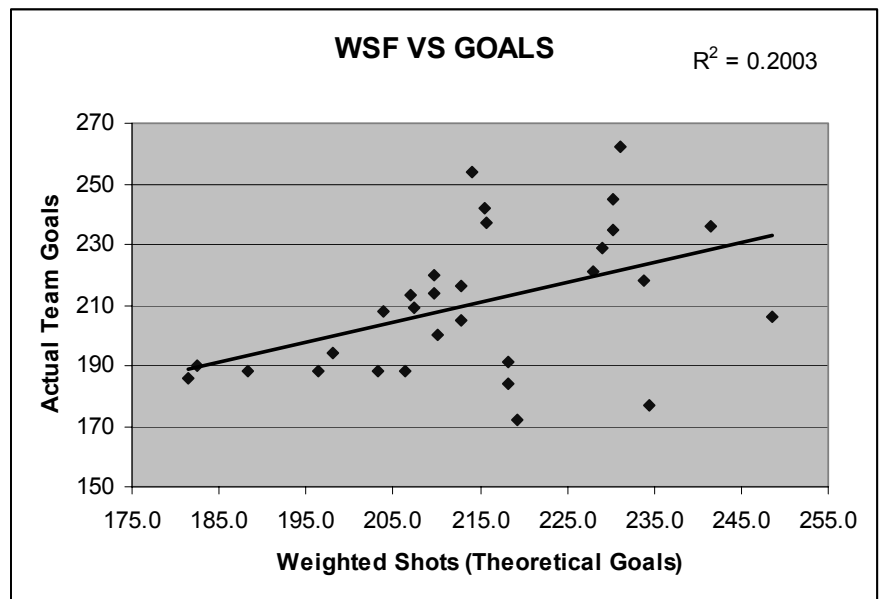
The talented forward Lecavalier took more of his shots from prime scoring locations than did the defender MacInnis.

Of course these results are not surprising, the positions of the players dictate the type and frequency of their respective shots. Lecavalier is expected to get more goals – he plays a primarily offensive role on the team – whereas MacInnis is not expected to produce his shots by penetrating the defence of the opposing team. In fact I am confident a significant percentage of MacInnis' shots are not taken with the intent of scoring, rather they are used to create higher percentage shots for the forwards in front of the other teams net by way of rebounds or tip-ins. This distinction is easy to observe when WSF is included alongside a players traditional offensive statistics.

Limitations of WSF

When considering this analysis on a team-by-team basis, the results become a bit fuzzy. Consider the following 2003-2004 regular season data:

TEAM	GOALS	SHOTS	WSF
ANA	184	2438	218.3
ATL	214	2141	209.6
BOS	209	2500	207.3
BUF	220	2301	209.6
CAL	200	2244	210.1
CAR	172	2394	219.3
CBJ	177	2280	234.5
CHI	188	2401	203.3
COL	236	2411	241.5
DAL	194	2197	198.1
DET	254	2482	214.0
EDM	221	2332	228.0
FLA	188	2273	196.4
LA	205	2349	212.8
MIN	188	1982	188.3
MON	208	2265	203.9
NAS	216	2227	212.8
NJD	213	2433	207.0
NYI	237	2341	215.7
NYR	206	2415	248.7
OTT	262	2421	231.2
PHI	229	2408	229.1
PHO	188	2160	206.4
PIT	190	2027	182.5
SJ	218	2295	233.7
STL	191	2223	218.2
TB	245	2460	230.3
TOR	242	2253	215.6
VAN	235	2379	230.3
WAS	186	1946	181.4



It seems that a combination of shots and shot quality (WSF) correlates poorly to overall team offence. So what we have is a few teams who are greatly under or over performing their expected goal scoring distribution. Is that luck? Skill?

Detroit seems to be the most obvious example. Based on their shot distribution, they had outscored the league average goal probabilities by an incredible 40 goals in 03-04. Similarly, Ottawa (30.8 goals), Tampa Bay (14.7), and Toronto (26.4) all overachieved their “expected goals for”.

There could be several reasons for this:

1. In many ways the results for those teams may not be surprising. There is a difference between shot quality as I have primitively defined it and shot execution, and common sense suggests that all four of these teams are loaded with “snipers”, who could possess a repeatable skill of outperforming the league average scoring probabilities. After all, isn’t that just another way of saying great player?

In hindsight, the WSF statistic assumes everyone in the league will have a scoring distribution that looks like the one in the graph on page 1. This is probably unrealistic, but it is done as a matter of convenience. That model is formed on the data from about 60000 shots per year, which allows a nice smooth distribution to form. With individual players noise is an unavoidable issue. It is likely that all players could eventually form their own shot probability logistic, which would differ from player to player, but players would need to play many more games than they do. Unfortunately with the available data speculation is all that I can offer at this point.

In a way, goal scoring can be thought of as a function of three significant variables:

1. Number of shots.
2. Quality of shots taken.
3. Shot execution.

WSF is appealing because it is an attempt to quantify the effects of both 1 and 2. Limitations of the shot quality model aside, any variations in shooting style should manifest itself within WSF. If a player is not maximizing efficiency between quantity and quality of his shots it will result in lower WSF totals.

Shot execution on the other hand is difficult to categorize. Traditional shooting percentage is one way of looking at it, but it is subject to problems, because it doesn’t account for shot quality. A better way would be to look at a players over/ under production of his WSF. If some players or teams are consistently scoring more goals than their WSF totals would predict, it would suggest that somehow that player or team is shooting more accurately, or otherwise able to produce higher percentage shots than the league average.

2. Perhaps these teams are overachieving somewhat, and their anomalous statistics are the result of some wealth of luck or unsustainable circumstance. If this is the case, we could expect some regression from those teams in the coming seasons (assuming similar rosters), at least in terms of their offensive production.

To further this thought, notice that on the offensive side a lot of truly terrible teams also outperformed theory, namely Atlanta (4.4), Washington (4.6), Pittsburgh (7.5). But worst of all, if the sniper theory is

true, how come other teams with premium scoring talent like Colorado(-5.5), Philadelphia(0.1), New York Rangers(-42.7), St. Louis(-27.2) do not even come statistically close to the Ottawa's and Toronto's of the NHL?

Seeing such a mess of numbers also made me doubt my judgement on the defensive and goaltending side. Perhaps I am too quick using the same play-by-play data to sing the praises of Dwayne Roloson, and maybe I was much too hard on Martin Brodeur or Tommy Salo.

The Shot Execution Gap

The fact that some teams are doing such unexpected things leads one to believe that there may be other forces at work here. There is a gap, a realm of uncertainty between the WSF predictions and actual results. For now I will refer to the discrepancy between theory and results as the "Shot Execution Gap (SEG)".

Assigning labels and causality to a gap in a statistical result is risky business. There can be many reasons for the SEG, and weeding out the true answer is almost impossible without allowing some sort of subjective bias to trump reason. Dismissing the SEG as luck is probably premature, but it may also be financially worth proceeding with caution with a player who has one-year goal totals that look way out of line with the rest of his career.

Here is a quick rundown of some possible explanations of the SEG:

1. Shot accuracy – shooting percentage
2. Player confidence, being "hot" or "cold"
3. The effects of play-making and efficient passing
4. Bounces or other intangible circumstance
5. Systematic errors in the play-by-play data

Factors 2, 4 and 5 are not really all that useful from a forecasting perspective. They do not really have any predictive value that is useful to gauge future performance. When a player will become hot, cold, lucky or unlucky is impossible to predict.

Explanations 1 and 3 may show up in the data if we look creatively. I have started this line of analysis, and while mathematically the research is still in relative infancy, I could find very little evidence of their existence. To elaborate, let's go back to Detroit in 03-04 and find out specifically which players were responsible for offensive over-production.

Detroit scored 40 more goals than my shot quality model would predict. So who on that team was responsible for such a discrepancy? Originally I would expect Hull, Shanahan, Yzerman, or others who are known for shot accuracy.

2003-2004 Season (goals / weighted shots/ career shooting percentage)

Brett Hull (**25** / **16.9** / 15.2%)
 Steve Yzerman (**18** / **14.8** / 15.0%)
 Pavel Datsyuk (**30** / **16.2** / 17.8%)

Okay, that accounts for about 25 of the 40 surplus goals, but it may surprise you where the other goals came from.

Kris Draper (**23** / **16.7** / 10.2%)
 Kirk Maltby (**14** / **9.5** / 10.2%)
 Tomas Holmstrom (**15** / **11.5** / 15.8%)
 Steve Thomas (**10** / **7.6** / 13.3%)

That is a list of pretty good hockey players, but not exactly a list of snipers. This is even more surprising when you consider some of the players that over-production *did not* come from.

Brendan Shanahan (**25** / **24.6** / 13.4%)
 Henrik Zetterberg (**15** / **18.0** / 13.6%)

Datsyuk's over-production is particularly alarming. Almost 14 goals! If Ryan Smyth shot as efficiently as that he would have well over 100 goals in the last 2 seasons. If we look at Datsyuk's traditional career shooting percentage:

YEAR	SHT%
2002	13.90
2003	14.60
2004	22.10

Datsyuk's 2003-2004 performance looks way out of line with the rest of his career. Unless this is an example of a superstar emerging, I would be very surprised if he can maintain his goal scoring numbers in the coming years. Luckily for Detroit, Datsyuk's good fortune equated to a lot of extra goals scored during the regular season. Perhaps we should not have been surprised by the offensive meltdown Detroit has experienced in the playoffs for the last couple of years...

Looking at some other year-to-year examples from other teams (goals / WSF)...

Havlat (OTT) Career Sht% - 15.2%
 2002-2003 (21 / 25.3)
 2003-2004 (31 / 22.9)

Same player in theory, better "luck" in 03-04? Assuming that he sustains no major injury, we should see about 25 goals or so from him next season.

White (OTT) – 14.4%
 2002-2003 (25 / 15.9)
 2003-2004 (9 / 13.1)

Got a lot of bounces in 02-03, again similar WSF numbers in 04 despite missing 20+ games, wildly different (potentially millions of dollars different) goal totals.

Marchant (EDM / CBJ) – 9.0%
 2002-2003 (20 / 15.7)
 2003-2004 (9 / 18.0)

Has a great year in Edmonton, gets signed as a free agent for a raise, and then regresses, to the disappointment of the Blue Jackets.

Teemu Selanne (2002-2004) – 15.4%
(38 / 52.6)

Much has been made about what a bust he was for Colorado, but defensive responsibilities aside, maybe it has more to do with bad luck. He still looks capable of being extremely productive.

The majority of players seem to hover around their theoretical goal value, but it seems that the actual goal scored total is often subject to large swings. Perhaps some players may have their value inflated by such swings; one career-best season could be the result of some statistical fluke. Conversely, and more importantly, some players may be undervalued because their ability to produce quality scoring chances is hidden by their disproportionately low goal totals.

Quantity of Chances or Execution of Shots?

If there is a causal explanation for the SEG beyond randomness, it may yet still show up in the data. To quickly have a look, I have made a couple of assumptions:

1. If there are players taking advantage of shot accuracy or efficient passing that could influence the SEG, their regular season goal totals will vary predictably from their WSF numbers. Is this happening?
2. Essentially an extension of assumption 1, if such phenomena are highly relevant to goal scoring, the same players will over/ under perform their WSF totals from year to year. Is this happening?

There are two observations that suggest the answer to 1 is no.

For individual players, 2002-2003 WSF correlates just as well ($R^2 = 0.451$) as 2002-2003 goals ($R^2 = 0.447$) with 2003-2004 regular season goal totals.

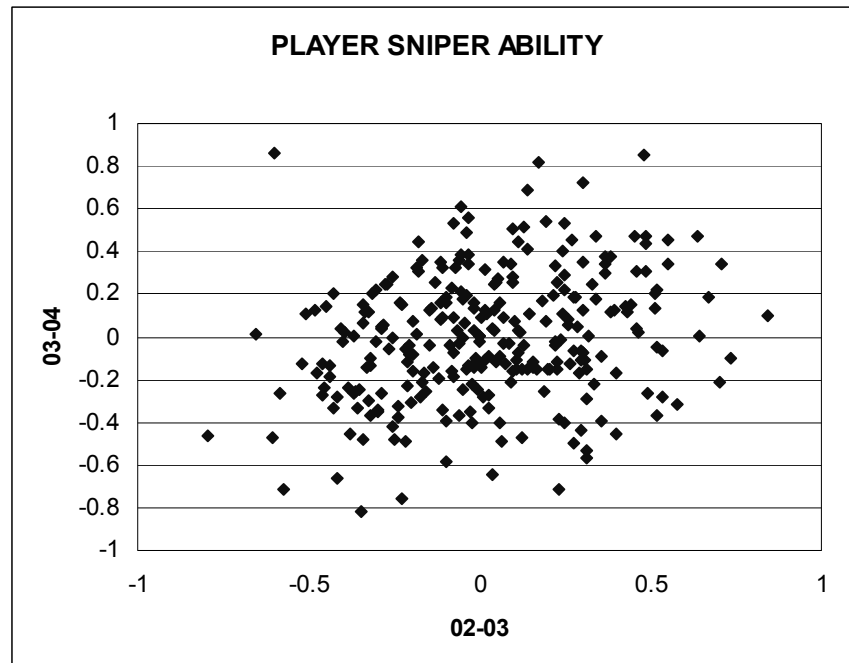
For individual players, WSF actually correlates better from year to year ($R^2=0.566$) than actual goals scored ($R^2 = 0.447$).

I would argue that this provides evidence that what makes an effective goal scorer at the NHL level is a player's ability to get shots from prime scoring locations, not necessarily to shoot accurately.

In response to the second assumption, a closer look reveals that there is little to no correlation of player over or under-achievement from season to season. To show this, I have defined over/under production as the rate of scoring in proportion to WSF.

In other words,

$$SNIPER\ ABILITY = \frac{GOALS - WSF}{WSF}$$



The definition is admittedly somewhat arbitrary, however the results are very interesting. The R^2 correlation of 0.058 is very weak; suggesting that over and under-production as defined above is not a sustainable trait for the overwhelming majority of NHL players.

If I were a Detroit or Toronto fan these results would greatly concern me. Not only are many of their offensive players expensive, ageing and fighting nagging injuries, but these results suggest that last year they may also have been performing at a rate they probably cannot expect to sustain.

The case for “Snipers”

To be honest, I am not convinced yet that shooting accuracy is merely a trivial factor in NHL goal scoring. I have my opinions, and possibly my biases, but in many ways the purpose of this paper is an attempt to initiate debate, hopefully spurring response from thoughtful people more statistically inclined than myself.

I outlined my concerns to Alan Ryder, who also has done some research on the topic, and he seems to defend the sniper theory.

I also reached the conclusion that offensive shot quality was poorly correlated to offense. I interpreted this to mean that shot quality was a question of style.

The variation from "expected" is either luck or execution or both. I think there is a lot of luck (round rubber thing on slippery surface) and I think that players get hot and cold.

I am more inclined to think that it is execution. Your list of "snipers" is a who's who on offence. Maybe these guys have been lucky. Maybe you make your luck. The list of laggards includes several players I consider overrated. They certainly haven't performed.

It is easy to dismiss it as random (= luck, =bounces). But the simple fact that, on this metric, some players are well over and some players are well under suggests that it is possible to out-perform the "model". I certainly hope this is possible! It would be tragic if ALL that mattered in shooting was positioning.

A limited number of players excel over their careers (Hull). They are consistent at a higher level. Others flame out. They can be white hot for a season or two and then are never the same. They might be explained by a fundamental law of statistics ("the Law of Very Large Numbers") that extreme performance will always arise.

The lists that Alan Ryder is referring to are shown below. The lists represent the top over and under achievers over the available data range (approximately 2 regular seasons).

TOP NHL UNDERACHIEVERS 02-04				TOP NHL OVERACHIEVERS 02-04			
PLAYER	WSF	GOALS	DIFF	PLAYER	WSF	GOALS	DIFF
25 DAMPHOUSSE	43.84	28	-15.84	17 HULL	40.49	60	19.51
15 PRIMEAU	29.73	14	-15.73	27 MURRAY	53.00	72	19.00
8 SELANNE	52.59	38	-14.59	15 HEATLEY	34.12	52	17.88
88 LINDROS	40.11	26	-14.11	18 HOSSA	60.37	78	17.63
27 TAYLOR	22.75	9	-13.75	13 DATSYUK	22.94	40	17.06
39 WEIGHT	36.24	25	-11.24	26 ST. LOUIS	48.03	65	16.97
24 KAPANEN	31.23	20	-11.23	23 HEJDUK	65.20	82	16.80
94 SMYTH	58.80	48	-10.80	38 DEMITRA	42.63	59	16.37
16 HOLIK	50.76	40	-10.76	12 BONDRA	32.18	47	14.82
25 PRIMEAU	33.50	23	-10.50	17 KOVALCHUK	58.77	73	14.23
12 STAAL	21.41	11	-10.41	20 LANG	35.61	49	13.39
19 MCEACHERN	33.40	23	-10.40	61 STILLMAN	31.94	45	13.06
20 DVORAK	35.26	25	-10.26	91 FEDOROV	48.15	61	12.85
17 CLARK	27.23	17	-10.23	19 NASLUND	65.22	78	12.78
71 HECHT	17.12	7	-10.12	10 MORROW	30.28	43	12.72
17 BRIND'AMOUR	29.60	20	-9.60	11 ALFREDSSON	42.81	55	12.19
8 SANDERSON	52.33	43	-9.33	13 SUNDIN	51.16	63	11.84
23 CHISTOV	21.06	12	-9.06	19 THORNTON	45.36	57	11.64
7 CLYMER	14.05	5	-9.05	7 ROBERTS	22.53	33	10.47
11 MADDEN	36.01	27	-9.01	16 LINDEN	22.54	33	10.46

Alan's points are certainly well taken. The over-achieving list is certainly a who's who on offence, despite some notable names missing from the list, Jagr, Kovalev, Yashin to name a few. The list of under-achievers is mostly just that: a group of disappointing, not to mention ageing, under-achievers. These results would seem to suggest that players would benefit greatly from increasing their shooting accuracy, and GM's should be wary of over-valuing players who get a lot of chances but have trouble finishing them.

A quick look at the non-adjusted shooting percentages of some of these players reinforces this position. Brett Hull is a great example. Hull takes a lot of shots, often at a significant distance away from the net. However, he has always been known for a quick release and remarkable accuracy. There are some small blips, but overall it looks like his shooting percentage is quite stable, and it is therefore probable that his goal scoring over-achievement is sustainable.

YEAR	SHT%
1998	12.80
1999	16.70
2000	10.80
2001	17.80
2002	12.10
2003	14.10
2004	12.50

I would argue that this is the intuitive explanation, and I believe Alan is right, at least for most of the extreme cases. If the play-by-play statistics recorded shot angle, speed, and other significant variables some of the discrepancies we see in the results could probably be attributed to some quantifiable, but yet unrecorded statistic. There are a multitude of intangibles in the shot quality model that could be statistically significant, such as the presence of a great passing line-mate, which would surely effect the quality of chances seen by some players given the same relative shot distances.

Disclaimers, Conclusions and Subjectivity

I must caution that the statistics that are calculated based on these play-by-play reports need to be treated with caution. I think much larger samples are needed before we can determine the real significance of both the defensive and offensive numbers. Alan Ryder explains in his Shot Quality paper how inaccurate the distances are, and sometimes it becomes difficult to remember that, but I believe it is essential to keep it in mind. Looking at systematic differences in the play-by-play statistics from rink to rink is also an important area of further research. If NHL teams are not already some kind of shot quality analysis, they should be, and they should start by collecting accurate data.

But let us pretend for a moment that these problems do not exist, that systematic errors are not a problem, and that we are forced to draw conclusions from this work.

Offensively, and to some extent defensively, I think these results are analogous to the defence independent pitching statistics that baseball analysts work on. More specifically, these results look remarkably similar to the relationship of pitchers and their ability to produce outs on balls in play. At first glance, looking only over a 2 year basis, the numbers suggested that pitchers have no control, the results are random with results wildly varied. However there is some evidence that they do have some very limited control, it is just overwhelmed by the noisy data/small sample size. A few rare pitchers can actually develop a small edge, and actually make a career out of getting more ground and fly balls for outs than others. But for most part, deviations from the average represent some kind of unsustainable run of luck.

Baseball analysts have explained how Ichiro Suzuki is one of the rare baseball players who has significant control over balls in play where most others statistically do not. And like baseball analysts, I

can certainly buy the argument that there are a few elite players who have made a career out of scoring from a distance. Brett Hull, and players of similar styles immediately come to mind, and I do not think it is a coincidence that he shows up at the top of the over-achievers list. In fact, that very list is more or less a who's who of players who are known for deadly power-play one-timers.

Keep in mind though that there is a substantial element of uncertainty involved, and GM's are forced to do a significant amount of guesswork into the future performance of their players. It appears that there is enough luck to drastically effect the result of an 82 game regular season, not to mention a short series in the playoffs. Based solely on this data, if the 2003-2004 season were to go on for another 82 games or so, I would not be surprised to see a significant offensive regression by Detroit, and to see huge improvements in the offensive output of the Rangers and Hurricanes.

From this perspective, perhaps one regular season is much too small of a sample size to reliably determine a player or teams true offensive ability. To put forward a subjective argument, I often find that goals seem very random in the modern game; goal-mouth scrums, deflections, screened shots, etc. Goalies do not really do a whole lot of reacting; they are wisely concentrating on becoming much more efficient with their positioning. I think that in the vast majority of circumstances, improved defence has forced players to not aim very precisely when they shoot... they do not have time. In retrospect that was often the case when I played.

There is an obvious hierarchy of NHL goal scoring talent, but the data suggests that what makes a good goal scorer at the NHL level is a player's ability to get shots from prime scoring locations, not necessarily to shoot accurately. WSF seems to be a good indicator of this ability, and when used in combination with players traditional stats, it provides a more complete picture of offensive ability.